21-242 Recitation 12

Span, Linear Independence, Bases — Q2. Definition of linear independence

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

State the definition of linear independence.

Course-level solution

Definition (Axler / sequential form)**.** A list $(\vec{v}_1, \ldots, \vec{v}_n)$ in V is *linearly independent* if no vector in the list is a linear combination of the preceding ones: that is, $\vec{v}_1 \neq \vec{0}$ and $\vec{v}_j \notin \operatorname{Span}(\vec{v}_1, \ldots, \vec{v}_{j-1})$ for each $j = 2, \ldots, n$. The empty list is linearly independent.

A list is *linearly dependent* if it is not linearly independent.
(An equivalent and often more convenient form is Q3 below.)